The L3 cognate faciliation effect

Is cognate facilitation **cumulative**?

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Introduction

Cognate faciliation refers to ease in processing cognates relative to non-cognate words.

There is robust evidence of this effect in L2 studies: when one word shares orthographic form in two languages, it is more easily processed (faster reaction times on lexical decisions).

Most models of word recognition suggest that increased cross-language overlap of form and meaning facilitates lexical activation and recognition (e.g., Dijkstra & Van Heuven, 2002; Dijkstra et al., 2018; Shook & Marian, 2013; Léwy & Grosjean, 2008).

Lemhöfer et al. (2004) and Szubko-Sitarek (2011) found cumulative cognate facilitation effects in trilinguals (L1 Dutch, L2 English, L3 German; and L1 Polish, L2 English, L3 German, respectively). The overlap of form and meaning across three languages yielded shorter response times than across two languages:
trilingual cognates < bilingual cognates < non-cognate words

Research Questions

RQ1: Do trilinguals process three-way cognates faster than two-way cognates?
RQ2: Is cognate faciliation modulated by L2 or L3 proficiency?

Methodology

Participants

L3 Spanish - L2 English - L1 German (n = 48)
L2 Spanish - L1 English (n = 49)

Experiment

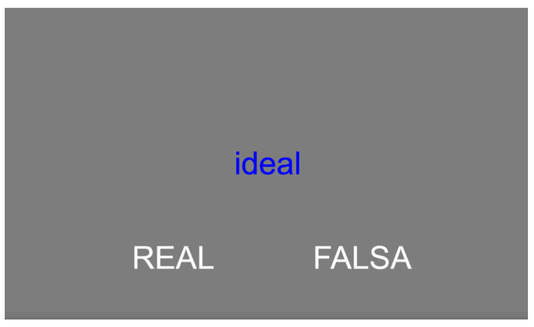

Figure 1.
Sample trial of lexical decision task.

Lexical Decision Task: Is this string of letters a word in Spanish?

Stimulus list: 192 items

- 96 Spanish Pseudowords
- 48 Non-cognate real words
- 24 Two-way identical cognates (English, Spanish)
- 24 Three-way identical cognates (English, Spanish, German)

Example stimuli:

- *Three-way cognate:* horror, horror, Horror
- *Two-way cognate:* piano, piano, Klavier
- *Non-cognate:* face, rostro, Gesicht

Results

Overall cognate effect? Yes.

Overall, cognates yielded faster responses (mean 870ms, sd 310 ms) than noncognates (mean 930ms, sd 320ms).

A Bayesian mixed effects regression found this difference compelling [- .047; 95% HDI -.032 - -.011; pd = .996] (see Figure 2).

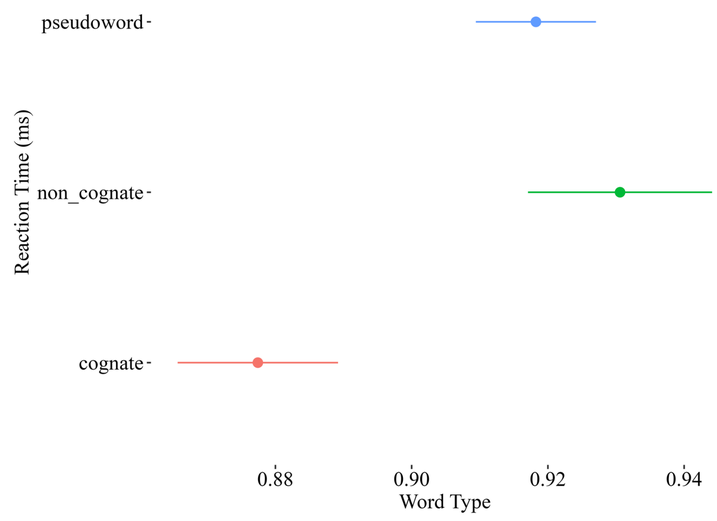

Figure 2.
Reaction times by word type in the L3 group.

Cumulative cognate effect? Inconclusive, but unlikely.

There was no evidence of a cumulative cognate facilitation effect [- .017; 95% HDI -.058 - .025; pd = .79; ROPE = 76%] (Figure 3).

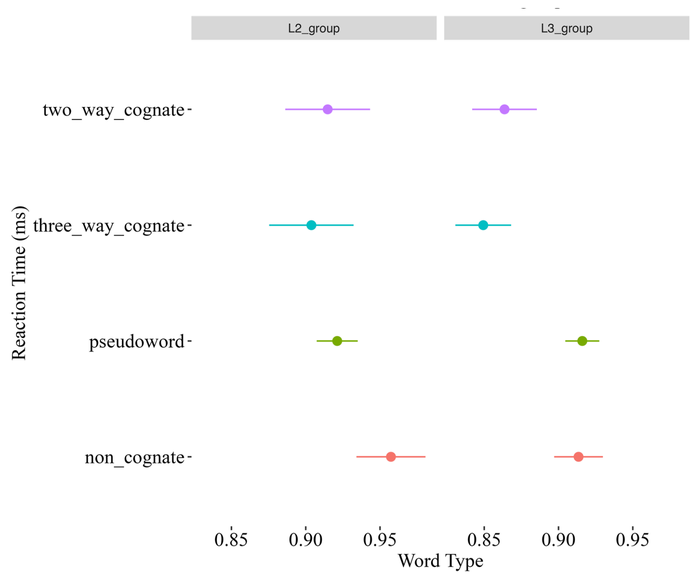

Figure 3.
Reaction times by word type in both groups.

Effect of proficiency? No

There was no clear role of proficiency in a processing advantage [ROPE = .99; pd = .81] (Figure 4).

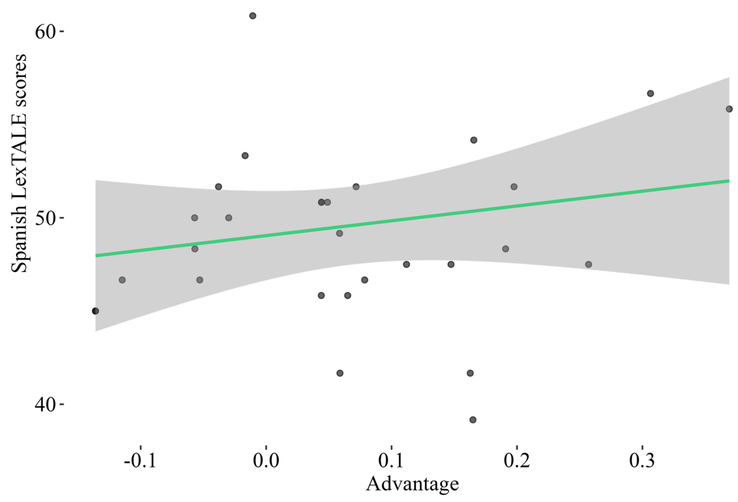

Figure 4.
Spanish Lextale scores compared to the cognate faciliation effect

Conclusions

- Consistent with previous findings, bilinguals and trilinguals processed cognates faster than noncognates, supporting nonselective lexical access and increased cross-linguistic coactivation of cognates.
- Unlike previous findings for trilinguals, overlap of form and meaning across three languages did not result in faster processing of L3 Spanish.
- L1 and L2 knowledge yielded similar facilitation effects on L3 lexical processing. The cumulative cognate effect reported in previous studies may depend on language type, task demands, and proficiency.
- L2 and L3 proficiency did not modulate processing of L2 and L3 Spanish cognates.